UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 011 (VoP) · Method of variation of parameters · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S011V·9	Unit 2 deck 011 “Variation of Parameters method” (SESSION 8), slide 9 (rendered .pptx)
T011V 00:03:26	AV-summary transcript “Variation of Parameters method” (14.9 min, listed first), segment start
T011Vb 00:16:53	AV-summary transcript “Variation of Parameters method_Continued” (24.5 min, second), segment start
CW U2 P25	Unit 1 & 2 class-work sheet, Unit 2 problem 25
ESA Jul-23 3c	Past paper: UE22MA141A July 2023, question 3.c (Jan-24 s1/s2 = UE23MA141A Jan 2024 sets 1/2)

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3. Figure G2.2 · $(D^2-1)y = \frac{2}{1+e^x}$: the PI follows $-X$ where X changes slowly
4. Figure G3.1 · ESA Jan-24 s2 3a: $y_p = \frac{1}{2}e^{-x}\sin x \tan x$
5. Figure G4.1 · ESA Jul-23 3c: $y_p = xe^{-3x}\log x$ and general solutions

The deck has no graphs, so every figure is added for clarity. Colours follow the Okabe–Ito order: blue = the particular integral, vermilion (dashed) = the right-hand side X, green and purple = the varying parameters $u(x)$, $v(x)$; grey for guides, asymptotes and sample general solutions. Every plotted function was checked by substituting it back with sympy.

1 The method: $y_p = uy_1 + vy_2$ with the Wronskian

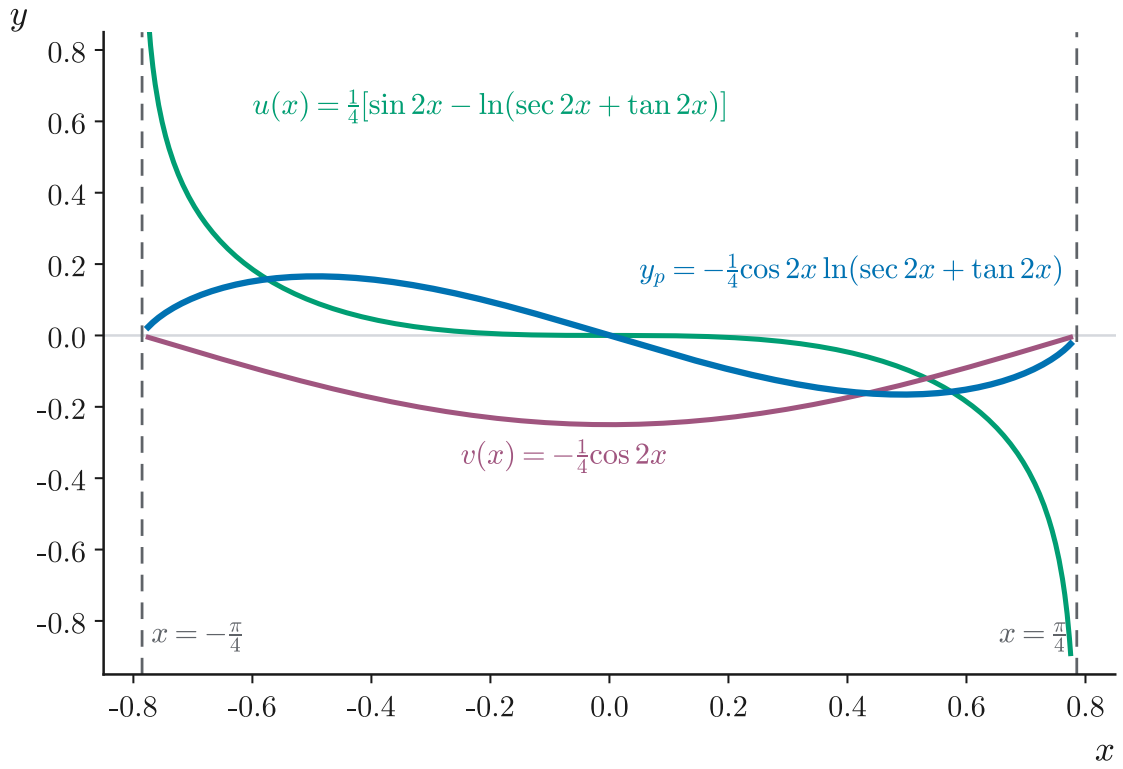


Figure G1.1. The parameters vary: $u(x)$, $v(x)$ and y_p for $(D^2 + 4)y = \tan 2x$. ESA ★★

ADDED FOR CLARITY S011V·12 · S011V·13

How to read it. On $-\frac{\pi}{4} < x < \frac{\pi}{4}$. Green: $u = \frac{1}{4}[\sin 2x - \log(\sec 2x + \tan 2x)]$; purple: $v = -\frac{\cos 2x}{4}$. In the CF they would be constants (horizontal lines); here they change with x, which is the idea of the method. Blue: $y_p = u\cos 2x + v\sin 2x = -\frac{1}{4}\cos 2x \log(\sec 2x + \tan 2x)$, which stays bounded although u blows up at the ends.

2 Slide and lecture examples: xe^{-x}, $\tan 2x$, $\frac{2}{1+e^x}$, a Cauchy equation

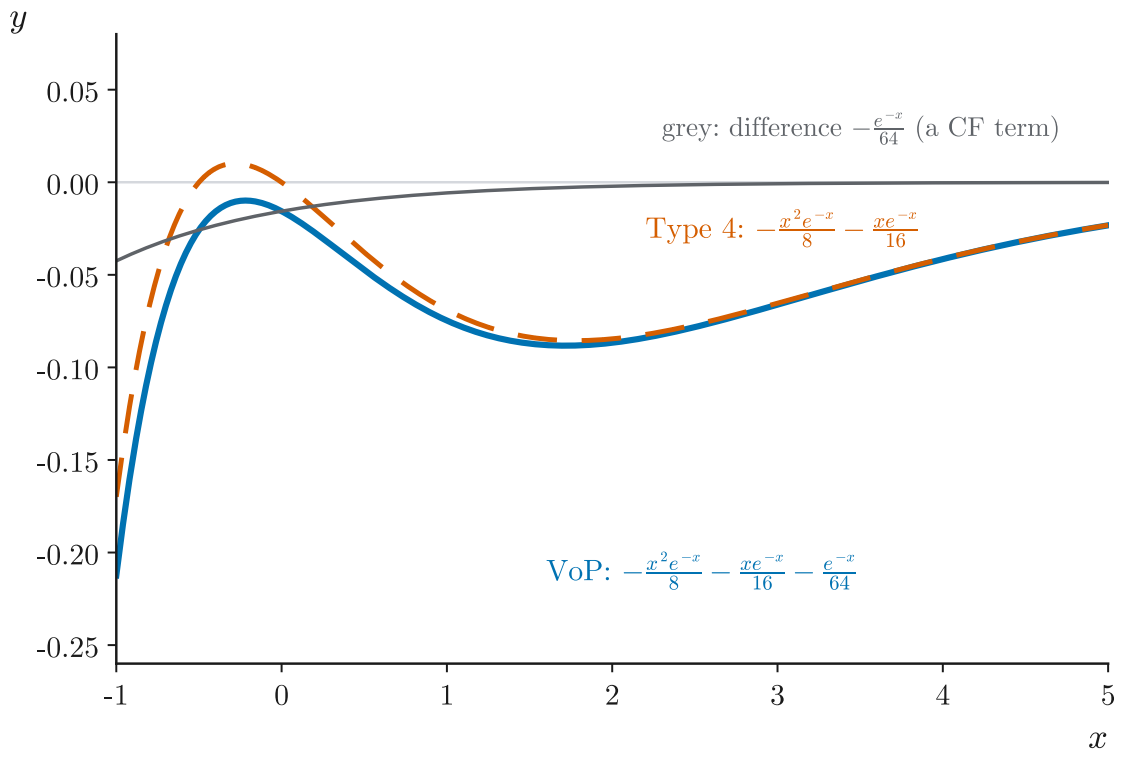


Figure G2.1. Slide 011V·10: two particular integrals differ by a CF term. ESA ★★

ADDED FOR CLARITY S011V·10 · T011V 00:08:02

How to read it. Blue: the VoP PI of slide 011V·10, $-\frac{x^2e^{-x}}{8} - \frac{xe^{-x}}{16} - \frac{e^{-x}}{64}$. Vermilion dashed: the Type 4 PI $-\frac{x^2e^{-x}}{8} - \frac{xe^{-x}}{16}$. Grey: their difference $-\frac{e^{-x}}{64}$, a multiple of $y_2 = e^{-x}$, so both are correct PIs.

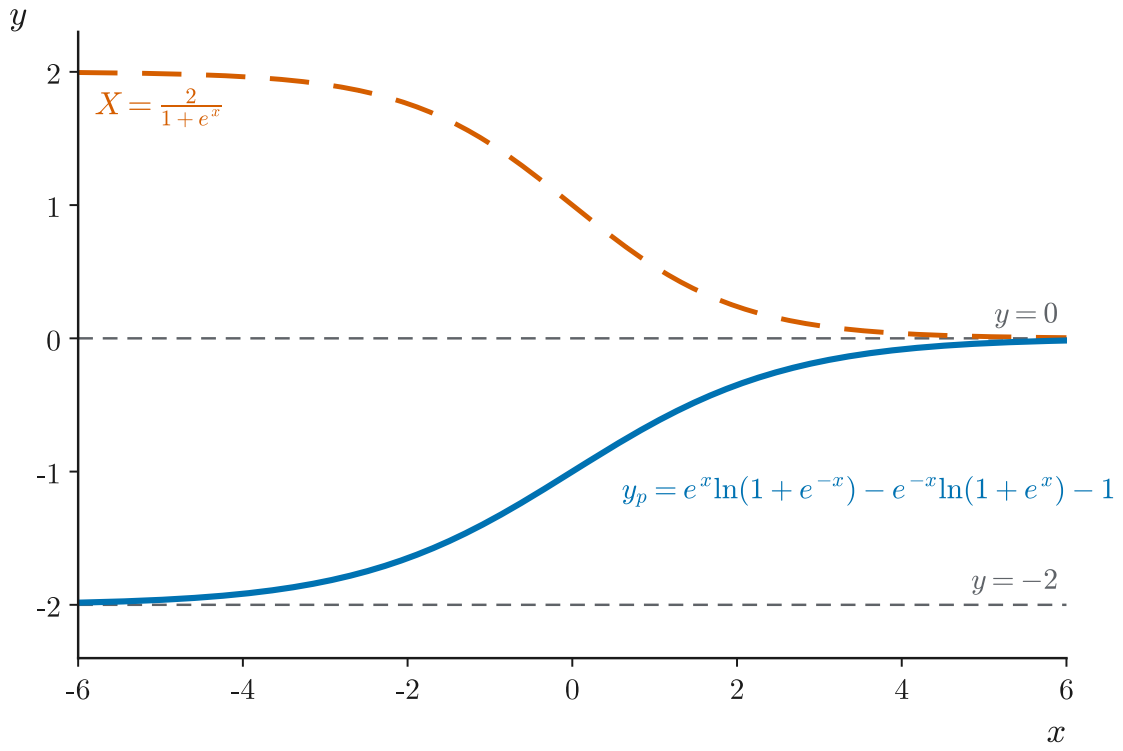


Figure G2.2. $(D^2-1)y = \frac{2}{1+e^x}$: the PI follows $-X$ where X changes slowly. ESA ★★

ADDED FOR CLARITY S011V·17

How to read it. Vermilion dashed: $X = \frac{2}{1+e^x}$, from 2 (left) to 0 (right). Blue: $y_p = e^x \log(1+e^{-x}) - e^{-x}\log(1+e^x) - 1$. For very negative x, $y_p \to -2$ (the PI of $y'' - y = 2$); for large x, $y_p \to 0$. Grey: the levels -2 and 0.

3 Trigonometric right-hand sides: $\sec ax$, $\csc x$, $e^{x}\tan x$, $e^{-x}\sec^{3} x$

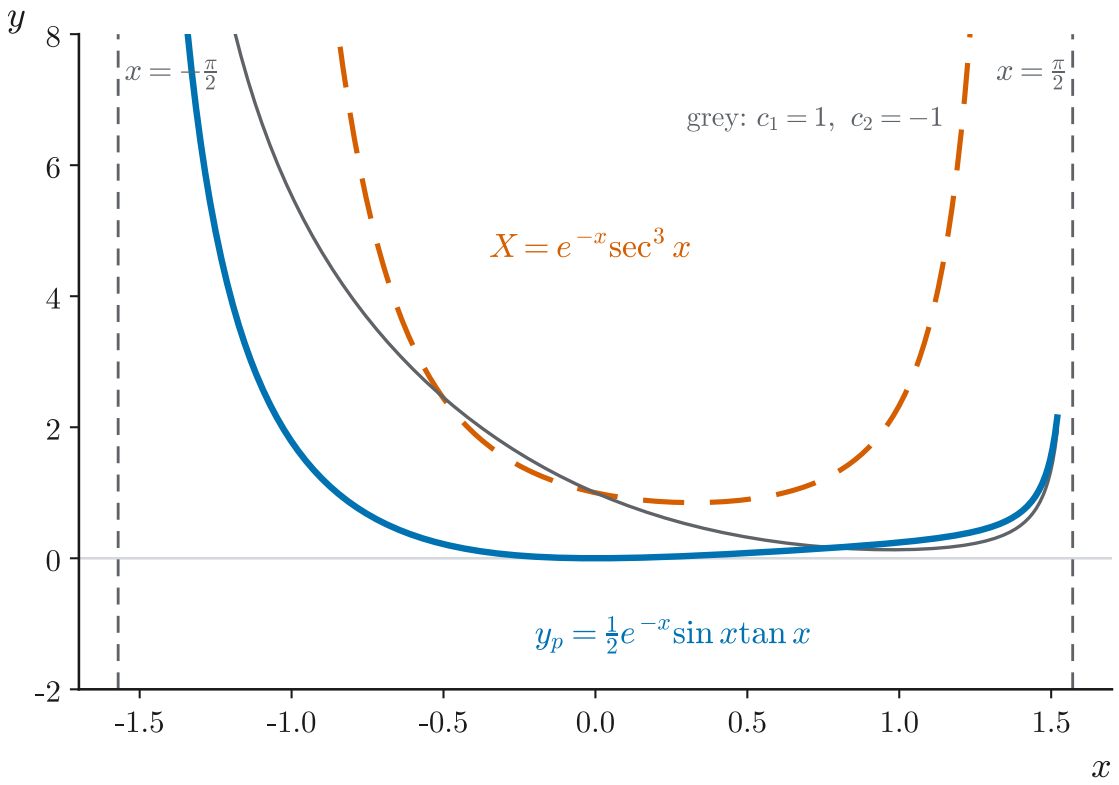


Figure G3.1. ESA Jan-24 s2 3a: $y_p=\frac{1}{2}e^{-x}\sin x\tan x$. ESA ★★★

ADDED FOR CLARITY ESA Jan-24 s2 3a

How to read it. On $-\frac{\pi}{2}<x<\frac{\pi}{2}$, where $\sec x$ exists. Vermilion dashed: $X=e^{-x}\sec^3 x$. Blue: the PI, zero at $x=0$ and unbounded at the asymptotes $x=\pm\frac{\pi}{2}$ (grey) like X. Grey solid: a general solution with $c_1=1$, $c_2=-1$.

4 Repeated roots with $X = \frac{e^{mx}}{x^k}$, and the leading-coefficient trap

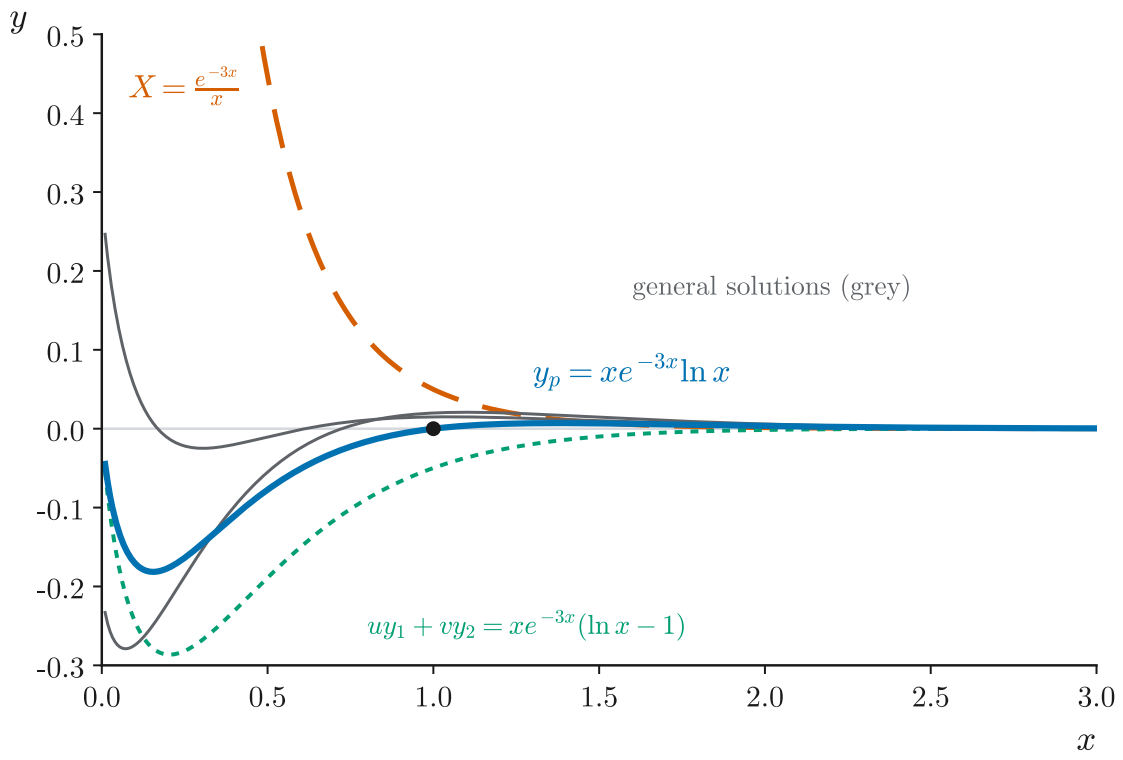


Figure G4.1. ESA Jul-23 3c: $y_p = xe^{-3x}\log x$ and general solutions. ESA ★★★

ADDED FOR CLARITY ESA Jul-23 3c · ESA Jun-24 3a

How to read it. For $x > 0$. Vermilion dashed: $X = \frac{e^{-3x}}{x}$, unbounded at $x = 0$. Blue: $y_p = xe^{-3x}\log x$, negative for $x < 1$, zero at $x = 1$, tending to 0. Grey: two general solutions $(c_1 + c_2 x)e^{-3x} + y_p$; the dotted green curve is the un-absorbed $uy_1 + vy_2 = xe^{-3x}(\log x - 1)$, itself one of the general solutions ($c_2 = -1$).